

Confluent Kafka Isotope

An example of using a Kafka **producer interceptor** to tag every message with a tracer — an *isotope* — that carries through every hop of a multi-topic pipeline. Consume-then-produce services adopt the inbound trace into thread-local with one explicit call (`IsotopeContext.adoptFromRecord(record)`) between consume and produce; the next `send()` then appends a fresh hop automatically. For full **bipartite-topology visibility** (services *and* terminal consumers), consumers also call `IsotopeContext.recordConsume(record, service, producer)` to emit a best-effort consume-edge marker to the `iso_consume_events` topic. Apache Flink consumes the tagged records and reports on what the isotopes reveal: **end-to-end latency**, **hop topology** (produce-side and the full **bipartite service↔topic↔service graph**), **drop/duplication rates**, and **pipeline coverage**.

A **portability requirement** runs through this project: the same isotope mechanism must work against both **Confluent Cloud for Apache Flink (CCAF)** (managed) and **Confluent Platform for Apache Flink** (self-managed) — both used here via Table API SQL plus uploaded UDF/PTF JARs (no DataStream code on either side). Three decisions follow from that: tagging happens in a Kafka **producer interceptor** (the one extension point both runtimes share via the broker), the on-wire **header** format is **JSON** (so Flink SQL can read the scalar fields with `CAST(headers[...] AS STRING)` and no UDF), and the optional stateful reports (`LatencyPercentilesUDAF`, `StuckTracePTF`) ship as a single JAR that registers identically on either runtime.

One asymmetry the runtimes don't share: **Flink-native SR-Protobuf**. CCAF supports SR-framed Protobuf as a Flink sink format via its topic catalog; Apache Flink open-source (the CP Flink runtime) ships `avro-confluent` but no SR-Protobuf counterpart. So Flink *report sinks* land on **Avro+SR on CP** and can be **Protobuf+SR on CCAF** — a runtime constraint, not a project preference. The demo *event* topics (next paragraph) are unaffected because they're written by the Kafka producer client, not by Flink.

Message **values** on the demo topics are **SR-framed Protobuf** (`ai.signalroom.kafka.isotope.proto.DemoEvent`) — the standard Confluent value format. The interceptors and reports are agnostic to value format because the isotope rides in headers; the Protobuf choice just gives the integration tests and the demo CLI a typed payload to work with.

Table of Contents

- [1.0 How the isotope is carried](#)
 - [2.0 Architecture](#)
 - [3.0 Repo layout](#)
 - [4.0 Running](#)
 - [4.1. Unit tests \(no broker, instant\)](#)
 - [4.2 Demo CLI — see one trace propagate live](#)
 - [4.3 Integration tests \(live Kafka via Minikube\)](#)
 - [4.4 Flink SQL reports on Confluent Platform for Apache Flink \(Minikube\)](#)
 - [4.4.1 Format-by-domain](#)
 - [4.5 Flink SQL reports on Confluent Cloud for Apache Flink \(CCAF\)](#)
 - [4.6 Recommended path the first time through](#)
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1.0 How the isotope is carried

- **Header `x-isotope`** (JSON bytes) carries the full hop history, forwarded by every hop:
 - `t` — 16-byte **UUIDv7** trace ID (RFC 9562): 48-bit ms timestamp in the high bits + 74 bits random. Stable for the life of the trace, and lexicographic byte order matches creation order — sort trace IDs and you get chronological order for free.
 - `o` — origin timestamp (ms) — same value as the timestamp embedded in the UUIDv7 trace ID; kept as its own field for typed access from Flink SQL without needing to decode the UUID bytes.
 - `s` — origin service name (set once, never reassigned)
 - `h` — ordered list of hops, each `{s: service, t: topic, m: tsMs}`
 - `x` — `true` if the hop list exceeded `MAX_HOPS = 32` and the oldest hop was evicted
- **Six scalar headers** (UTF-8 strings) carry the most-recent-hop view so Flink SQL can read them via `CAST(headers['x-isotope-...'] AS STRING)` without parsing the JSON array (no UDF required on either CCAF or CP Flink). See <scripts/flink/README.md> for the full header table.

A producer with the isotope interceptor loaded appends one hop on every `send()`. A consume-then-produce service calls `IsotopeContext.adoptFromRecord(record)` between consume and produce so the trace ID and origin survive the hop.

How the producer interceptor gets invoked. Application code never calls `onSend` / `onAcknowledgement` directly — the Kafka client invokes them by reflection once two things are in place:

1. **Register the class in producer config.** Put `IsotopeProducerInterceptor.class.getName()` under `interceptor.classes` ([App.java:185](#), [App.java:250](#), [IsotopeTestHarness.java:96](#)). The `KafkaProducer` constructor instantiates the interceptor and owns its lifecycle.
2. **Call `producer.send(...)` as normal.** Every `send()` triggers `onSend` (caller thread, before serialization) and later `onAcknowledgement` (producer I/O thread, after broker ack or failure).

The exact call sites in `kafka-clients 4.2.0`: `KafkaProducer.send` invokes `interceptors.onSend` ([line 950](#)) and `AppendCallbacks.onCompletion` invokes `interceptors.onAcknowledgement` ([line 1600](#)). Exceptions thrown by the interceptor are caught and logged by the client's fan-out wrapper — they never break the `send` call.

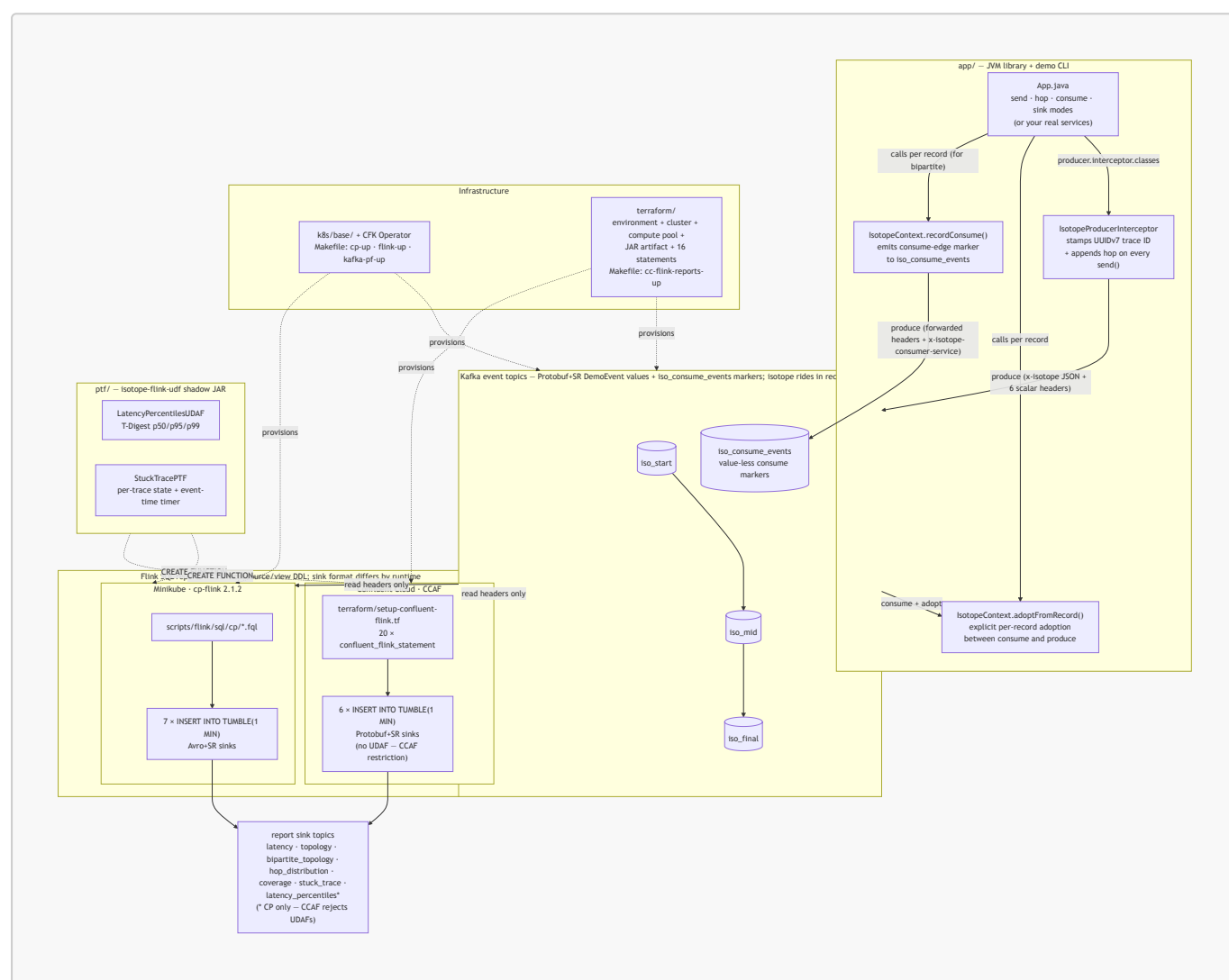
Consumer-side adoption is explicit, not an interceptor. A `ConsumerInterceptor.onConsume` sees a whole batch from `poll()`, but the application processes records one at a time, so a thread-local snapshot from the batch would be ambiguous. Instead, services call `IsotopeContext.adoptFromRecord(record)` per record between consume and produce. The next `send()` then sees that thread-local and appends a new hop. This project ships no consumer interceptor at all — the producer interceptor plus per-record adoption is the entire **produce-side** mechanism.

Consume-side markers (optional, for bipartite topology). The produce-side `hops[]` chain encodes the consume edges for intermediate services (svc-B consuming from topic-AB is implied by svc-B's next produced hop), but **terminal consumers** — services that consume but never produce — leave no trace and are invisible to the topology report. To make them visible, consumers call `IsotopeContext.recordConsume(record, service, markerProducer)` between consume and process. The call forwards the inbound record's six scalar `x-isotope-*` headers to a value-less marker record on `iso_consume_events`, adding one new header — `x-isotope-consumer-service` — that names the consumer. The bipartite-topology Flink report unions these markers with the existing produce-

side view to emit edges in both directions: **producer** → **topic** (from **isotope**) and **topic** → **consumer** (from **consume_events**). Markers are fire-and-forget: a dropped marker leaves a hole in the topology graph but never disrupts the consume/produce pipeline.

2.0 Architecture

A bird's-eye view of the moving parts. The JVM library in **app/** registers a Kafka producer interceptor that stamps the isotope into record headers on every **send()**; consume-then-produce services adopt the inbound trace via an explicit **IsotopeContext.adoptFromRecord(record)** call; records flow through a 3-topic chain; Flink SQL reads only the headers and emits 1-minute aggregate reports. The same source/view DDL deploys to both runtimes — **CP** on Minikube applies **.fql** files under **scripts/flink/sql/cp/**, and **CCAF** in Confluent Cloud applies inline **confluent_flink_statement** resources under **terraform/**. The shadow JAR from **ptf/** (which powers two of the six reports) registers identically on both. (Kafka is drawn once below for brevity — each runtime provisions its own cluster.)



See § 3.0 for the file tree behind each box, and § 4.0 for the run commands.

3.0 Repo layout

app/
tests

isotope JVM library + demo CLI +

src/main/proto/ai/signalroom/kafka/isotope/proto/

demo_event.proto	DemoEvent message (Protobuf value schema)
src/main/java/ai/signalroom/kafka/isotope/Isotope.java	POJO + JSON codec + Hop +
fromHeaders()	+ UUIDv7 helpers (uuidV7Bytes /
uuidV7String)	
IsotopeContext.java	ThreadLocal + adoptFromRecord() + recordConsume() (emits consume-edge
markers)	
IsotopeProducerInterceptor.java	stamps/appends x-isotope + 6 scalar reporting headers on send()
App.java	demo CLI – send / hop / consume /
sink modes	
src/test/java/.../IsotopeContextRecordConsumeTest	IsotopeCodecTest,
	(no broker needed)
src/integrationTest/java/.../ProducerInterceptorIT,	BrokerSmokeIT,
	ThreeStageHopPropagationIT,
BipartiteTopologyIT,	IsotopeTestHarness – live-broker
tests; produce/consume	
	DemoEvent via SR-framed Protobuf (need Minikube CP + SR port-
forwarded)	
ptf/	Flink PTF + UDAF shadow JAR (powers
2 of 6 reports)	
src/main/java/ai/signalroom/kafka/isotope/flink/	
LatencyPercentilesUDAF.java	T-Digest p50/p95/p99 aggregate
StuckTracePTF.java	per-trace state + event-time timer
Percentiles.java	p50/p95/p99 return-type DT0
src/test/java/.../	LatencyPercentilesUDAFTest
k8s/base/	CFK manifests
confluent-platform-c3++.yaml	Kafka / SR / Connect / ksqlDB /
Control Center	
flink-basic-deployment.yaml	cp-flink session cluster + CMF
flink-rbac.yaml	RBAC for the cp-flink operator
scripts/	
port-forward-kafka.sh	localhost:30092 → Kafka,
localhost:8081 → SR	
port-forward-taskmanager.sh	Flink TaskManager web UI forward
deploy-cp-flink-reports.sh	builds shadow JAR + applies
sql/cp/*.fql to	
	the cp-flink session cluster
deploy-cc-flink-reports.sh	builds shadow JAR + wraps
`terraform apply`	
	for the CCAF path
cc-cli-env.sh	pulls Kafka + SR creds from
`terraform output`,	
	builds the JAAS string, exports
BOOTSTRAP /	
	SR_URL / KAFKA_KEY / KAFKA_SECRET /
JAAS / ...	

```

cc-app-run.sh
:app:run` that

the six -D flags
  flink/README.md
(CP=6 reports/Avro+SR,

layout, operations
  flink/sql/cp/
01_register_functions,

06_consume_events_view,

reports, 99_teardown

terraform/setup-confluent-flink.tf.)
terraform/
cc-flink-reports-up`)
  providers.tf
key/secret vars
  versions.tf
provider versions
  variables.tf
region, day_count
  data.tf
sources
  setup-confluent-environment.tf
governance package)
  setup-confluent-kafka.tf
rotation module

tf_module)
  setup-confluent-flink.tf
compute pool,

rotation, and 20 inline

resources: 4 ALTER TABLE

CREATE FUNCTION

UDAFs) + 6 INSERT INTO
  outputs.tf
rotating

(sensitive)
  terraform.png
in § 4.5)
Makefile
flink-reports-up /

reports-down / ...

```

```

thin wrapper around `./gradlew

sources cc-cli-env.sh and injects

Flink SQL reports – runtime split

CCAF=5 reports/Protobuf+SR),

CP Flink SQL: 00_source_table,

05_isotope_view,

05_report_sinks (avro-confluent),
10/20/25/30/40/60/70 INSERT INTO

(CCAF SQL is inlined under

CCAF infrastructure-as-code (`make

Confluent provider – cloud

required Terraform (>= 1.13) +

confluent_api_key/secret, cloud,

organization lookup + other data

environment (ESSENTIALS stream-

Kafka cluster + Kafka API key

(iac-confluent-api_key_rotation-

service account + 6 role bindings,

artifact upload, SR API key

`confluent_flink_statement`

+ 3 VIEW + 6 sink CREATE TABLE + 1

(PTF; UDAF skipped – CCAF rejects

environment_id, bootstrap, SR URL,

Kafka + SR API key/secret outputs

rendered resource graph (embedded

cp-up / flink-up / kafka-pf-up /

cc-flink-reports-up / cc-flink-

```

4.0 Running

4.1. Unit tests (no broker, instant)

```
./gradlew test # both subprojects
# or scoped:
./gradlew :app:test # IsotopeCodecTest (10) +
IsotopeContextRecordConsumeTest (4) – JSON roundtrip, hop eviction, UUIDv7
properties, consume-marker emission (14 tests)
./gradlew :ptf:test # LatencyPercentilesUDAFTest – T-
Digest accumulator semantics
```

4.2 Demo CLI — see one trace propagate live

The fastest way to watch the isotope mechanic. Requires the cluster to be up and the Kafka + SR forwards running (see step 3 below for the bring-up commands). The CLI has four modes:

Mode	Args	What it does
send	<topic> <service> <payload>	Produces one isotope-tagged DemoEvent to <topic>, then exits. Auto-creates the topic.
hop	<in-topic> <out-topic> <service>	Consumes records from <in-topic>, adopts the isotope into thread-local, emits a consume-edge marker to iso_consume_events as <service> , then re-produces the same DemoEvent to <out-topic> (which appends a new hop). Runs until Ctrl-C.
consume	<topic> <service>	Terminal-consumer mode for bipartite-topology demos. Subscribes to <topic>, emits a consume-edge marker to iso_consume_events as <service> , and pretty-prints the isotope trail. No downstream produce. Runs until Ctrl-C.
sink	<topic>	Passive peek tool — pretty-prints the isotope trail but does NOT emit a consume marker. Use for ad-hoc inspection; use consume to make a node visible in the bipartite-topology report. Runs until Ctrl-C.

A 4-stage chain (full bipartite graph) in five terminals:

```
# Terminal A – terminal consumer (will print the full 3-hop trail AND emit
a
# consume-edge marker so svc-D shows up in the bipartite
report)
./gradlew :app:run --args="consume iso_final svc-D" -q

# Terminal B – middle stage: iso_mid → iso_final as svc-C
./gradlew :app:run --args="hop iso_mid iso_final svc-C" -q
```

```
# Terminal C – first stage: iso_start → iso_mid as svc-B
./gradlew :app:run --args="hop iso_start iso_mid svc-B" -q

# Terminal D – kick the chain off (run repeatedly to send more)
./gradlew :app:run --args="send iso_start svc-A 'hello world'" -q
```

Terminal A's output for each record shows the same `trace_id` across all three hops, `origin = svc-A` (never reassigned), and `hops[]` listing `svc-A → svc-B → svc-C` in order with per-hop timestamps. The bipartite-topology report sees all six edges: produce edges `svc-A → iso_start`, `svc-B → iso_mid`, `svc-C → iso_final` and consume edges `iso_start → svc-B`, `iso_mid → svc-C`, `iso_final → svc-D`. Swap `consume` for `sink` if you only want to inspect records without recording the terminal edge. Override endpoints via `-Dkafka.bootstrap=...` / `-Dschema.registry.url=...` if you're not on the default Minikube layout.

4.3 Integration tests (live Kafka via Minikube)

Bring up the local Confluent Platform stack and port-forward Kafka + SR:

```
make minikube-start          # one-time
make cp-up                   # CFK Operator +
Kafka/SR/Connect/ksqlDB/C3 (~5 min)
make kafka-pf-up             # localhost:30092 → Kafka,
                              localhost:8081 → Schema Registry
```

Then run the suite:

```
./gradlew :app:integrationTest #
every IT below
./gradlew :app:integrationTest --tests '*ProducerInterceptorIT' #
just one
```

Override the endpoints if needed:

```
./gradlew :app:integrationTest \
  -PkafkaBootstrap=localhost:30092 \
  -PschemaRegistryUrl=http://localhost:8081
```

Tear down forwards when done:

```
make kafka-pf-down
```

The integration tests cover:

Test	What it verifies
<code>BrokerSmokeIT</code>	AdminClient can create/list/delete a topic via the NodePort port-forward
<code>ProducerInterceptorIT</code>	A consumer sees the <code>x-isotope</code> JSON header + all 6 scalar reporting headers with the expected origin/hop values, and the Protobuf round-trip preserves <code>DemoEvent.source / payload</code>
<code>ThreeStageHopPropagationIT</code>	<code>svc-A → topic-AB → svc-B → topic-BC → svc-C</code> produces a stable trace ID, 2-hop trail in send order, and correct scalar headers (origin = <code>svc-A</code> , this = <code>svc-B</code> , hop count = 2) at the terminal; consume-then-produce hops use <code>IsotopeContext.adaptFromRecord</code> to carry the trace forward
<code>BipartiteTopologyIT</code>	The 4-stage <code>svc-A → topic-AB → svc-B → topic-BC → svc-C → topic-CD → svc-D</code> chain emits exactly three consume-edge markers to a per-test markers topic — one per consume edge. Every marker carries the trace ID, forwarded <code>x-isotope-*</code> scalars describing the upstream producer, and the new <code>x-isotope-consumer-service</code> naming the downstream consumer. Asserts the <code>(consumer_service, consumed_topic)</code> set is exactly <code>{(svc-B, topic-AB), (svc-C, topic-BC), (svc-D, topic-CD)}</code>

4.4 Flink SQL reports on Confluent Platform for Apache Flink (Minikube)

The five pure-SQL reports plus the two JAR-backed reports (one PTF, one UDAF) — seven in total — run against a Flink session cluster managed by the Confluent Flink Kubernetes Operator. Same FQL files deploy to Confluent Cloud for Apache Flink — see § 4.5 for that path; this section is the local-Minikube path.

Bring up Flink:

```
make flink-up          # cert-manager → CFK Flink Operator → CMF →
session cluster        # (~5 min the first time)
```

Deploy the reports — all 7 reports run on the `cp-flink` session cluster (Flink 2.1.2). Sink topics use Apache Flink's `avro-confluent` format — SR-framed Avro, auto-registered on first write — so Control Center renders the report rows natively.

```
make flink-reports-up
```

`flink-reports-up` builds the PTF/UDAF shadow JAR if missing, copies it into the JobManager pod, pre-creates the 7 sink Kafka topics, then applies the source + view + sink DDL and submits 7 `INSERT`

INTO streaming jobs (one per report). On first write to each sink, Apache Flink's `flink-sql-avro-confluent-registry` format registers a fresh Avro schema in SR under subject `<topic>-value`. **Control Center deserializes all 7 report topics natively** — no `.proto` files in the repo for the reports, no hand-installed format jars beyond the one init-container download.

4.4.1 Format-by-domain

The demo *event* topics (`iso_start`, `iso_mid`, `iso_final`) still ride **Protobuf+SR** via the Java app's `DemoEvent` schema — that's unchanged. The consume-edge marker topic `iso_consume_events` has **null value + scalar headers only** (Flink reads the headers via a `MAP<STRING, BYTES>` virtual column; there's nothing to deserialize). The *report* topics ride **Avro+SR** because cp-flink doesn't ship an SR-integrated Protobuf format and CMF (which does) disallows the UDAFs the percentiles report needs. Events from the app are Protobuf; aggregates from Flink are Avro; consume markers are headers-only. Format by domain — a clean split, not a defect.

4.5 Flink SQL reports on Confluent Cloud for Apache Flink (CCAF)

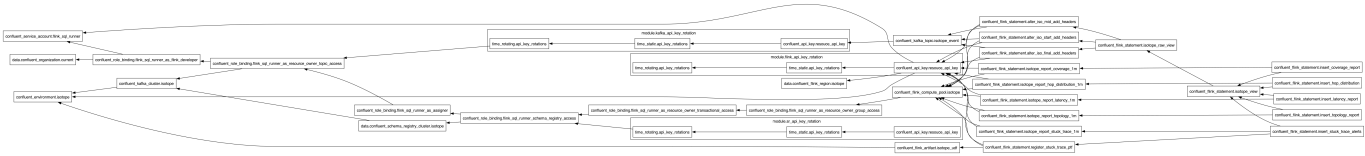
CCAF parallel of § 4.4, driven by Terraform under `terraform/`. One `make` target spins up a fresh Confluent Cloud environment, Kafka cluster, 4 pre-created event topics (the three demo topics `iso_start` / `iso_mid` / `iso_final` plus the consume-edge marker topic `iso_consume_events`; report sink topics are created on the fly by the Flink `CREATE TABLE` statements — see the comment in `terraform/setup-confluent-kafka.tf` for why pre-creating them via `confluent_kafka_topic` would conflict with CCAF's Topic Catalog auto-import), a Flink compute pool, two rotating service-account API key pairs (one for Kafka, one for Schema Registry), the PTF/UDAF JAR uploaded as a Flink artifact, and 20 long-lived `confluent_flink_statement` resources broken down as **4 ALTER TABLE** (add scalar headers on the event topics) + **3 VIEW** (raw + typed produce + typed consume) + **6 sink CREATE TABLE** + **1 CREATE FUNCTION** (PTF only — the UDAF is intentionally skipped, see the limitation note below) + **6 INSERT INTO** streaming jobs. The Terraform shape mirrors `apache_flink-kickstarter-ii` — same provider version, same `iac-confluent-api_key_rotation-tf_module`, same DROP-then-CREATE statement pattern.

Prereqs:

- `Terraform` `>= 1.13` installed locally.
- A Confluent Cloud API key (Cloud-level, not cluster-scoped) with permissions to create environments, Kafka clusters, Flink compute pools, service accounts, role bindings, Flink artifacts, and statements. Generate via Console → Settings → Cloud API keys.

Deploy:

```
export CONFLUENT_API_KEY=...
export CONFLUENT_API_SECRET=...
make cc-flink-reports-up CONFLUENT_API_KEY=$CONFLUENT_API_KEY
CONFLUENT_API_SECRET=$CONFLUENT_API_SECRET
```



The wrapper script ([scripts/deploy-cc-flink-reports.sh](#)) builds the PTF/UDAF shadow JAR if missing, then runs `terraform apply -auto-approve` in [terraform/](#). First-run takes ~6–8 minutes (Kafka cluster provisioning dominates). Re-applies are idempotent — `CREATE ... IF NOT EXISTS` plus `lifecycle { ignore_changes = [compute_pool] }` on every statement.

What gets created (see [terraform/setup-confluent-flink.tf](#) for the full graph):

Resource	Name	Notes
<code>confluent_environment</code>	<code>confluent-kafka-isotope</code>	ESSENTIALS stream-governance package
<code>confluent_kafka_cluster</code>	<code>kafka-isotope</code>	Standard, single-zone, AWS us-east-1 by default
<code>confluent_kafka_topic</code> × 4	<code>iso_{start,mid,final}</code> + <code>iso_consume_events</code>	Only the event topics + the consume-marker topic are pre-created. The 6 <code>isotope_report*_1m</code> sink topics are created on first deploy by their <code>CREATE TABLE</code> statement (CCAF's Topic Catalog auto-imports any pre-existing topic as <code>(key BYTES, val BYTES)</code> , which would silently no-op the typed <code>CREATE TABLE</code>). <code>terraform destroy</code> cleans them up via the environment cascade.
<code>confluent_service_account</code> + 6 role bindings	<code>isotope-flink-sql-runner</code>	FlinkDeveloper (org) + ResourceOwner on topic=* / transactional-id=* / group=* / SR subject=* + Assigner on the service account
<code>confluent_flink_compute_pool</code>	<code>isotope-flink-statement-runner</code>	10 CFU; headroom for 6 INSERTs + ad-hoc SELECTs
<code>confluent_flink_artifact</code>	<code>isotope-flink-udf</code>	Uploads <code>ptf/build/libs/isotope-flink-udf.jar</code>
<code>confluent_flink_statement</code> × 20	(see file)	4 ALTER TABLE (event-topic scalar headers) + 3 VIEW (raw + typed produce + typed

Resource	Name	Notes
		consume) + 6 sink CREATE TABLE + 1 CREATE FUNCTION (PTF only; UDAF skipped) + 6 INSERT INTO

Useful outputs:

```

terraform -chdir=terraform output environment_id
terraform -chdir=terraform output kafka_bootstrap_servers
terraform -chdir=terraform output schema_registry_url
terraform -chdir=terraform output -raw kafka_api_key      # sensitive
terraform -chdir=terraform output -raw kafka_api_secret  # sensitive

```

Format-by-runtime (not -by-domain). CP's reports land on **Avro+SR** (`'value.format' = 'avro-confluent'` in [scripts/flink/sql/cp/05_report_sinks.fql](#)). CCAF's reports land on **Protobuf+SR** (`'value.format' = 'proto-registry'` in each sink's WITH clause in [terraform/setup-confluent-flink.tf](#)). The two runtimes' SQL is otherwise unshared: CP's lives hardcoded in [scripts/flink/sql/cp/](#), CCAF's lives inline as `confluent_flink_statement` resources in [terraform/setup-confluent-flink.tf](#).

CCAF UDAF limitation. CCAF currently rejects all `CREATE FUNCTION` statements for user-defined aggregate functions ("aggregate functions are not supported"). The `LATENCY_PERCENTILES` UDAF therefore deploys on CP only; the percentile report (`latency_percentiles_flat_1m`) does not exist on CCAF. The other JAR-backed function, `STUCK_TRACE_PTF` (a `ProcessTableFunction`, not a UDAF), works on both runtimes. The JAR itself is portable — `LatencyPercentilesUDAF` ships with a `byte[]`-based accumulator and is ready to register when Confluent lifts the restriction. The six CCAF reports are: `latency` (avg/min/max), `topology` (produce-side), `bipartite_topology` (full service↔topic↔service graph), `hop_distribution`, `coverage`, `stuck_trace`.

Driving traffic — the 4-stage demo against CCAF. [App.java](#) reads four optional `-D` properties (`kafka.security.protocol`, `kafka.sasl.mechanism`, `kafka.sasl.jaas.config`, `schema.registry.basic.auth.user.info`) that default to plaintext-no-auth for Minikube. [scripts/cc-cli-env.sh](#) pulls the Kafka + Schema-Registry credentials from `terraform output` (both keys are rotated by `module.kafka_api_key_rotation` and `module.sr_api_key_rotation` in [terraform/setup-confluent-kafka.tf](#)) and builds the JAAS string.

The thin wrapper [scripts/cc-app-run.sh](#) sources the env helper then invokes `./gradlew :app:run` with the six `-D` flags — pass it the same `send` / `hop` / `consume` / `sink` args you'd give `App.java`:

```
# Four terminals (same A/B/C/D order as § 4.2). No manual env exports –
# the wrapper sources cc-cli-env.sh, which pulls everything from terraform.
scripts/cc-app-run.sh consume iso_final svc-D      # A – terminal consumer
                                                    (emits marker)
scripts/cc-app-run.sh hop iso_mid iso_final svc-C   # B
scripts/cc-app-run.sh hop iso_start iso_mid svc-B  # C
scripts/cc-app-run.sh send iso_start svc-A 'hello' # D
```

The wrapper hard-fails with a clear message if any of the seven required values is missing, so you'll never silently hand gradle empty `-D` values.

Terminal A prints the same `trace_id` across all three hops, and the CCAF report INSERTs populate as you fire Terminal D — `SELECT * FROM isotope_report_latency_1m`, `SELECT * FROM isotope_report_bipartite_topology_1m`, etc. in the Cloud Console SQL workspace. The bipartite report shows all six edges of the chain (3 produce + 3 consume).

Sustained traffic — required to see report rows. The six INSERT INTO jobs aggregate over `TUMBLE(event_time, INTERVAL '1' MINUTE)` windows, and a tumbling window only emits when the watermark advances past `window_end`. A handful of records burst from Terminal D within a single 1-minute interval will sit in one open window forever (the most-recent record is the watermark, and it never gets older than itself). Spread traffic across **multiple** windows so the watermark crosses each boundary:

```
# 30 records spaced 5s apart ≈ 2.5 minutes of event-time → spans 3+ windows
for i in {1..30}; do
  scripts/cc-app-run.sh send iso_start svc-A "burst-$i"
  sleep 5
done
```

Wait ~90 seconds after the *last* record before checking `isotope_report_latency_1m` (and friends) — that's the watermark catching up. The `stuck_trace_alerts_1m` sink only fires for traces that go ≥60s of event time without a fresh hop, so the burst above won't trigger it (every trace gets one record and ends — no stalled in-flight state). To exercise `STUCK_TRACE_PTF`: send a single record to `iso_start` and don't run the `svc-B` / `svc-C` hops, then keep sending unrelated records elsewhere so the watermark advances past the stuck trace's `event_time + 60s`.

Teardown:

```
make cc-flink-reports-down CONFLUENT_API_KEY=$CONFLUENT_API_KEY
CONFLUENT_API_SECRET=$CONFLUENT_API_SECRET
```

Runs `terraform destroy -auto-approve` — deletes every resource above, including the environment itself. Safe to run repeatedly.

4.6 Recommended path the first time through

1. `./gradlew test` — proves the codec + UDAF logic without any cluster.
2. `make cp-up && make kafka-pf-up && ./gradlew :app:integrationTest` — proves the broker + SR + interceptor + Protobuf path end-to-end.
3. The 3-stage demo CLI walkthrough above — visually shows the trace accumulating hops.
4. `make flink-up && make flink-reports-up && make flink-sql` — reports populate as you drive traffic via the demo CLI (see § 4.2).
5. (Optional) `make cc-flink-reports-up` — the CCAF parallel; see § 4.5 for prereqs and the SASL-config caveat for the demo CLI.